

Information Processing and Retrieval

UA.DETI.PRI - 2025/26

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The course

- ❖ Scientific area
 - Informatics
- ❖ Weekly schooling:
 - 3 hours of TP classes
- ❖ ECTS credits: 6
- ❖ Code: 43621

- ❖ Motivation



Objectives

- ❖ By the end of this course, you should be able to:
- ❖ Analyse and implement methods for text preprocessing, representation, and indexing.
- ❖ Apply and compare statistical, probabilistic, and neural language models, including large-scale models.
- ❖ Design and evaluate information retrieval and question answering systems using both classical and neural approaches.
- ❖ Develop and assess solutions for document classification, sentiment analysis, information extraction, and summarisation tasks.

Program

- ❖ C1. Text processing and representation
 - Tokenisation, filtering, and normalisation of terms
 - Sparse models; feature extraction
 - Distributed and contextualised representations
- ❖ C2. Language models
 - Boolean model and n-grams
 - Probabilistic language model
 - Recurrent neural network models
 - Large-scale language models (LLMs)
- ❖ C3. Information retrieval and automatic question answering
 - Text indexing, Neural models, Automatic question answering
- ❖ C4. Document classification and information extraction
 - Sentiment analysis, Named entity recognition, Relation extraction, Summarisation, Evaluation

Main bibliography

- ❖ Speech and Language Processing (3rd Edition): Daniel Jurafsky, James Martin, 2025,
 - <https://web.stanford.edu/~jurafsky/slp3/>
- ❖ An Introduction to Neural Information Retrieval, B. Mitra, N Craswell, 2018.
 - <https://www.microsoft.com/en-us/research/uploads/prod/2017/06/fntir2018-neuralir-mitra.pdf>
- ❖ Introduction to Information Retrieval, C.D. Manning, P. Raghaven, H. Schütze, 2008, Cambridge University Press.
 - <https://nlp.stanford.edu/IR-book/information-retrieval-book.html>

Evaluation

❖ Practical evaluation (AP) – 70%

– Assignment 1 (40%)

- 16-Oct Assignment 1 – Interim Presentation
- 13-Nov Assignment 1 – Final Presentation
- ... and continuous assessment

– Assignment 2 (30%)

- 04-Dec Assignment 2 – Interim Presentation
- 18-Dec Assignment 2 – Final Presentation
- ... and continuous assessment

– Submissions on the same dates on GitHub Classroom

❖ Theoretical-Practical evaluation (ATP) – 30%

– Exams/Quizzes – aligned with the end of each Assignment

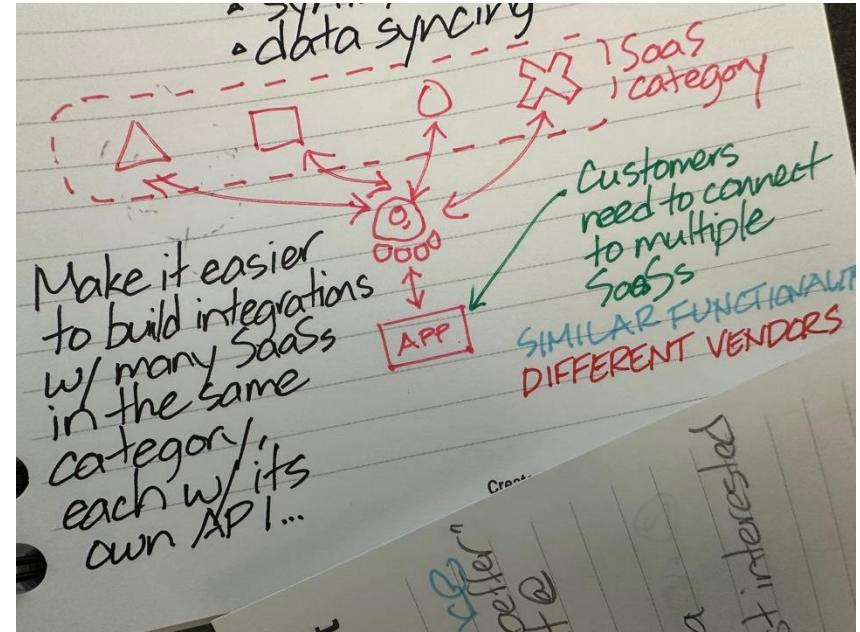
- 13-Nov Quiz 1 (20%)
- 18-Dec Quiz 2 (10%)

❖ Minimum score for each component: 7 points.

Notebook

- ❖ By the end of the semester, you have the option to submit a handwritten notebook with a collection of your own notes and takeaways from the lectures and assignments.

- Extra 5% of the final grade



ECTS

- ❖ Education (T/TP/P): 0/2/2 - ECTS: 6
- ❖ The number of ECTS credits indicates the expected number of hours they must study for this subject.
 - 1 ECTS = 25-30 hours of study.
 - 6 ECTS = 150-180 hours of study.
- ❖ In a 15-week semester, they must study at least 10 hours per week.
- ❖ These hours include face-to-face classes, reading books, solving exercises, studying for tests and exams, etc.

Resources

❖ elearning.ua.pt

- Slides
- Assignments (1 and 2)
- Submission instructions
- Information and results

❖ GitHub classroom

❖ Public resources

- Books, blogs
- Hugging Face
- Kaggle
- ... And many others

Teachers

- ❖ José Luis Oliveira (jlo@ua.pt)
 - IEETA
- ❖ Tiago Almeida(tiagomeloalmeida@ua.pt)
 - IEETA

Bons estudos e bom semestre!

